

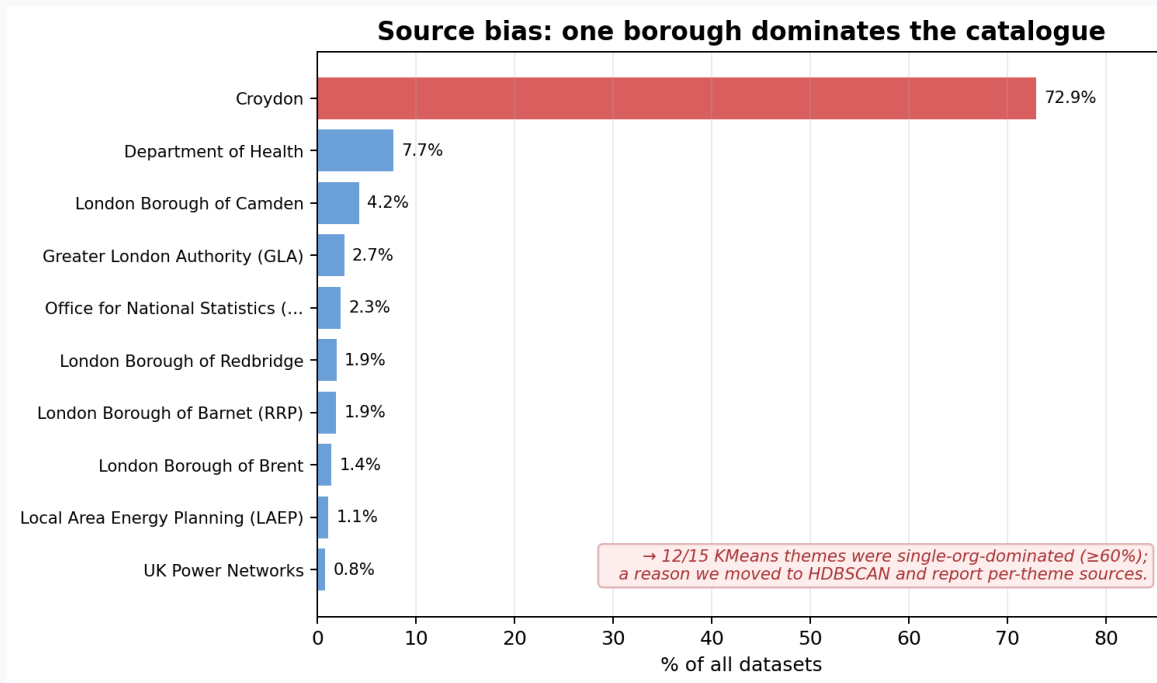
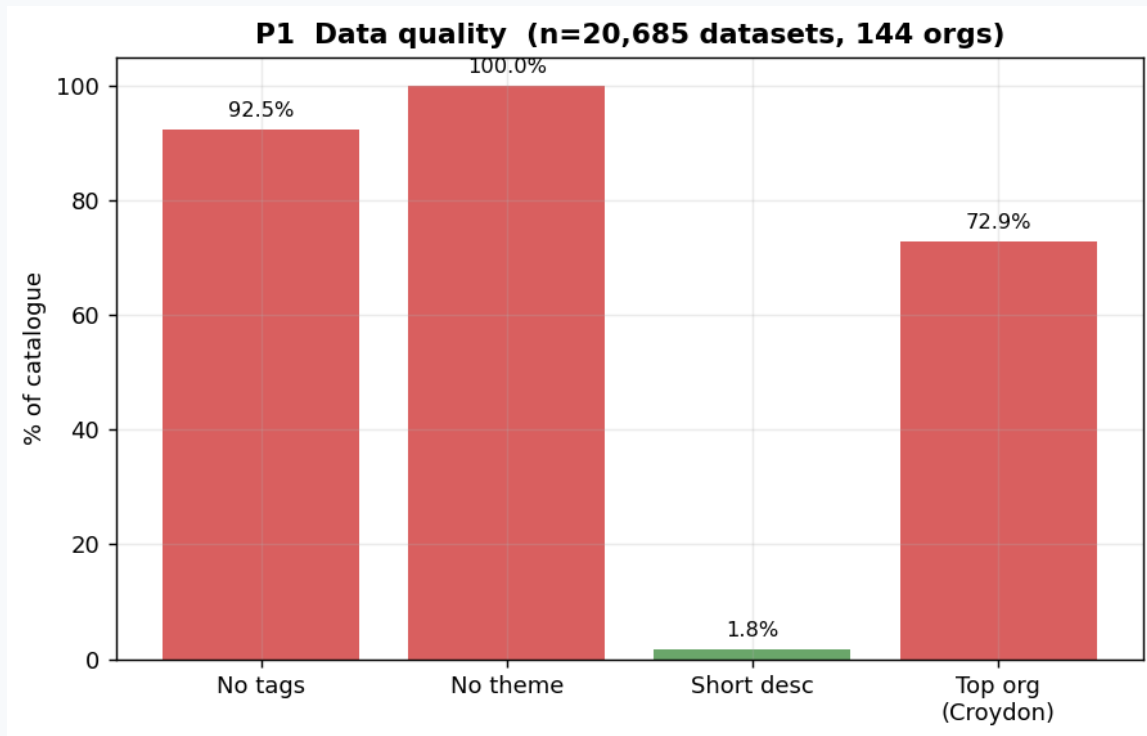
Can You Vouch For It?

Evaluating an Artificial Intelligence (AI) enrichment pipeline for 20,685 London datasets —
honestly.

Team: BY2 Xinrui QI & Yuyue Xia

Problem 3 · Discovering London's Data

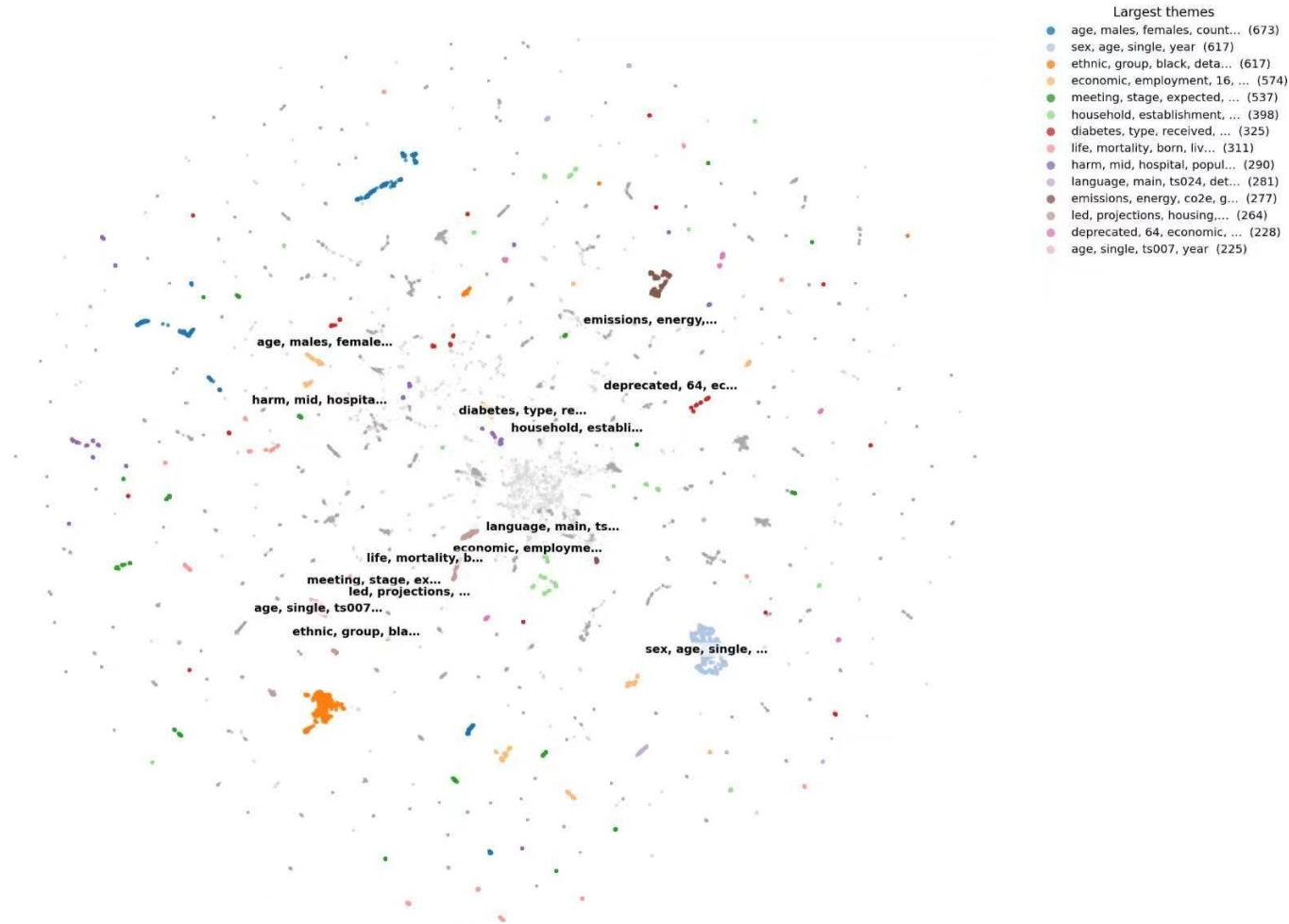
Good Data, Buried Deep



20,685 datasets. 92.5% have no tags, 100% have no theme, and one borough (Croydon) supplies 72.9% of everything — so today's search is really just a search of Croydon.

Turning Text Into Meaning

London's 20,685 datasets, organised by meaning
MiniLM embeddings → UMAP 2-D · coloured by top 14 of 76 discovered themes (grey = smaller themes / unclustered)



76

themes discovered automatically — no k
chosen by hand

How it works

Each dataset's title + notes become a 384-dimension MiniLM vector. Similar meaning = nearby vectors. HDBSCAN groups the dense areas into themes — grey dots are honestly left unclustered.

Search quality: two baselines vs. our method

Same query · same corpus (title + notes) · Top-5 results · 20,685 London datasets

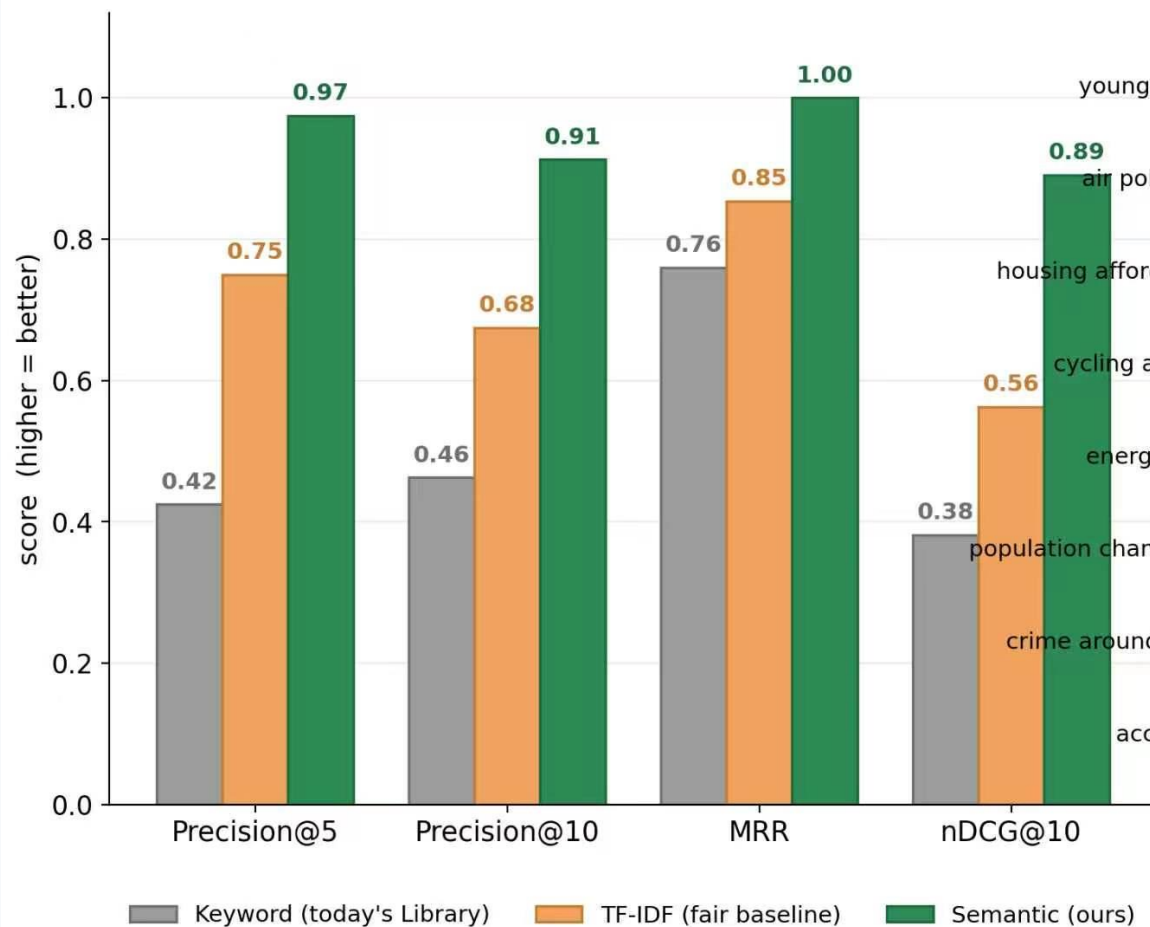
Query: "young people not in work" Synonym / acronym recall (NEET)		
Keyword Today's Library · word-count	TF-IDF Fair lexical baseline	Semantic Ours · MiniLM embeddings
1 London Labour Market, Skills and Employment Indicators ~	1 Young People ~	1 Percentage of male 16 and 17 year olds not in education, employment or training... ✓
2 TS058 - Distance travelled to work Works mainly from home rate ✗	2 Children & Young People ~	2 Number of male 16 and 17 year olds not in education, employment or training... ✓
3 TS058 - Distance travelled to work Works mainly from home ✗	3 Young people who have attained a full level 3 qualification by age 19 ~	3 Percentage of female 16 and 17 year olds not in education, employment or training... ✓
4 TS058 - Distance travelled to work Works mainly at an offshore installation, in... ✗	4 New analysis on autism in Camden: children and young people ~	4 Number of female 16 and 17 year olds not in education, employment or training... ✓
5 33kV Circuit Operational Data Half Hourly - South Eastern Power Networks... ✗	5 Children And Young People With Special Education Needs And Disabilities ~	5 16 to 17 year olds not in education, employment or training (NEET) or whose... ✓
✓ Relevant ~ Partial match ✗ Off-topic		
Query: "air quality and pollution" Robustness to long-text noise (COVID)		
Keyword Today's Library · word-count	TF-IDF Fair lexical baseline	Semantic Ours · MiniLM embeddings
1 Coronavirus (COVID-19) Vaccine Roll Out ✗	1 Camden Air Quality Monitoring ✓	1 Analysing Air Pollution Exposure in London ✓
2 Coronavirus (COVID-19) Deaths ✗	2 Analysing Air Pollution Exposure in London ✓	2 Air pollution: estimated fraction of mortality attributable to particulate... ✓
3 London Atmospheric Emissions Inventory (LAEI) 2019 ✓	3 London Air Quality Network Camden ✓	3 Air pollution: fine particulate matter (new method - concentrations of total... ✓
4 Coronavirus (COVID-19) Weekly Update ✗	4 London Air Quality Network Camden Latest Week ✓	4 Understanding Health Impacts of Air Pollution in London ✓
5 Public Health Outcomes Framework Indicators ~	5 Low Emission Neighbourhoods ✓	5 Fraction of mortality attributable to particulate air pollution - Persons ~... ✓

Keyword mimics today's Library · TF-IDF is a fair lexical baseline · embeddings recover synonyms (NEET) and resist long-text noise (COVID)

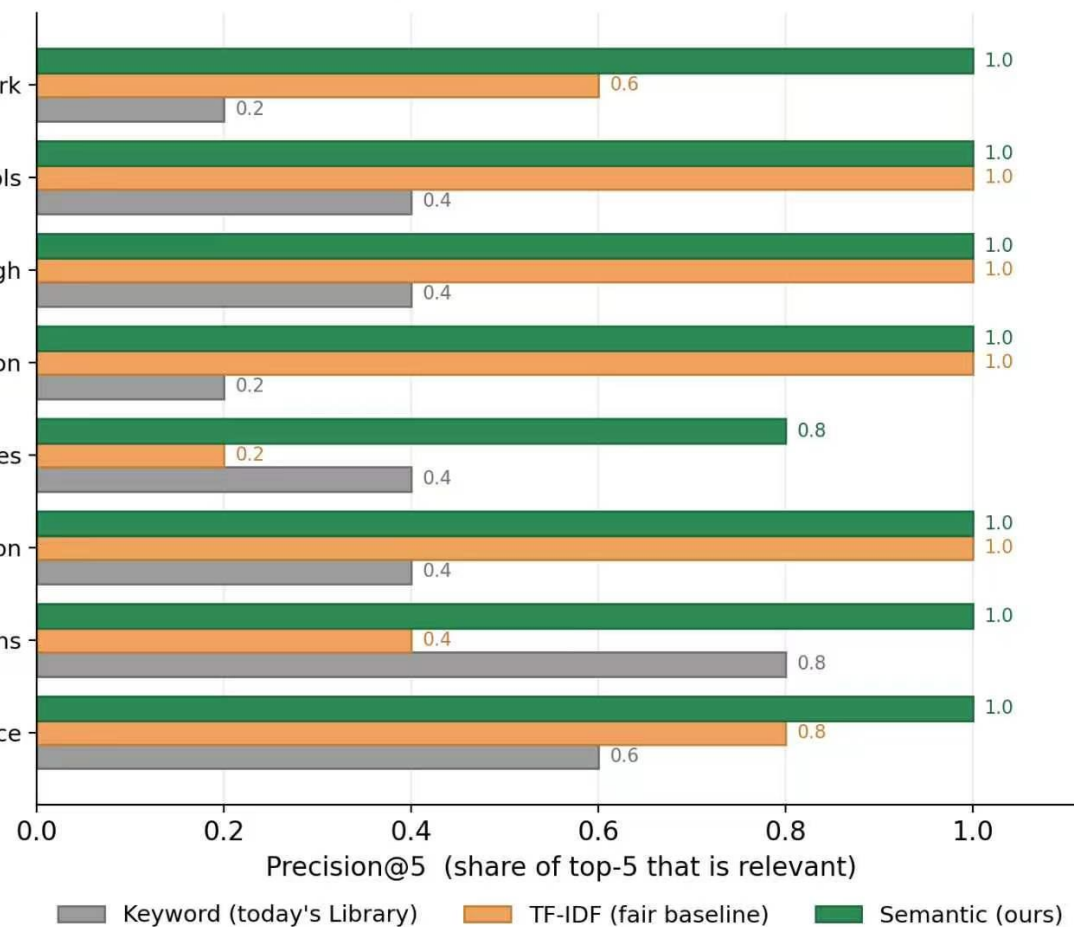
4 · MODEL EVALUATION · SEARCH (THE HEADLINE)

Semantic search evaluation — 8 natural-language policy queries, human-judged relevance (0/1/2), pooled top-10

Semantic search wins on every metric



Precision@5 holds across every query



Are the Auto-Generated Tags Any Good?

We manually reviewed a random sample of 30 datasets and their auto-tags.

93.3%

of sampled tags
were specific and
usable as
suggestions

15.8%

of all tags contain
stray digits (e.g.
'32', '500') —
noise to filter

7.7%

of tags are
purely numeric
— should be
dropped
automatically

1.43

average words
per tag — tags
run a little too
short/fragmented

dataset_idx	title	notes, snippet	tag	heuristic	human
108	Workflow Diagram: Buildings: Retrofit	These workflow diagrams show the data inputs, processing steps, assumptions and o retrofit	Good	Good	Good
108	Workflow Diagram: Buildings: Retrofit	These workflow diagrams show the data inputs, processing steps, assumptions and o workflow	Good	Good	Good
108	Workflow Diagram: Buildings: Retrofit	These workflow diagrams show the data inputs, processing steps, assumptions and o buildings	Good	Good	Good
108	Workflow Diagram: Buildings: Retrofit	These workflow diagrams show the data inputs, processing steps, assumptions and o workflow diagram	Good	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. TI number eligible	Good	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. TI eligible pupils	Good	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. TI additional	Good	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. TI english	Good	Good	Good
1147	DEPRECATED: Number of eligible pupils with English as additional language (EAL)	Note from Data for London: Croydon Observatory provided access to this dataset. TI eligible	Good	Good	Good
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. TI language	Good	Good	Good
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. TI language eu	Good	Good	Good
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. TI european language	Good	Good	Good
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. TI services	Okay	Okay	Okay
2461	Other European language (EU): Lithuanian %	Note from Data for London: Croydon Observatory provided access to this dataset. TI european	Good	Good	Good
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. TI housing led	Okay	Okay	Okay
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. TI led	Okay	Okay	Okay
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. TI 75 84	Bad	Bad	Bad
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. TI population projections	Good	Good	Good
2717	Males aged 75 to 84 %	Note from Data for London: Croydon Observatory provided access to this dataset. TI projections	Okay	Okay	Okay
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. TI	18	Bad	Bad
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. TI 18 years	Good	Good	Good
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. TI aged 18	Good	Good	Good
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. TI years persons	Good	Good	Good
4653	% aged under 18 years - Persons - <18 yrs (Upper 99.8% Confidence Limit)	Note from Data for London: Croydon Observatory provided access to this dataset. TI persons 18	Good	Good	Good
5270	AF006: stroke risk assessed w. CHAZD52-VASc (den.incl.exc) - Persons - All ages (Denominator)	Note from Data for London: Croydon Observatory provided access to this dataset. TI vasc	Good	Good	Good
5270	AF006: stroke risk assessed w. CHAZD52-VASc (den.incl.exc) - Persons - All ages (Denominator)	Note from Data for London: Croydon Observatory provided access to this dataset. TI risk	Good	Good	Good
5270	AF006: stroke risk assessed w. CHAZD52-VASc (den.incl.exc) - Persons - All ages (Denominator)	Note from Data for London: Croydon Observatory provided access to this dataset. TI stroke	Good	Good	Good
5270	AF006: stroke risk assessed w. CHAZD52-VASc (den.incl.exc) - Persons - All ages (Denominator)	Note from Data for London: Croydon Observatory provided access to this dataset. TI assessed	Good	Good	Good
5270	AF006: stroke risk assessed w. CHAZD52-VASc (den.incl.exc) - Persons - All ages (Denominator)	Note from Data for London: Croydon Observatory provided access to this dataset. TI ages denominator	Good	Good	Good
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. TI	59	Bad	Bad
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. TI population	Good	Good	Good
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. TI population estimates	Good	Good	Good
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. TI aged 16	Good	Good	Good
5757	Population aged 16-59	Note from Data for London: Croydon Observatory provided access to this dataset. TI population aged	Good	Good	Good
5891	DEPRECATED: Smoking status at time of delivery - Female - All ages	Note from Data for London: Croydon Observatory provided access to this dataset. TI smoking status	Good	Good	Good
5891	DEPRECATED: Smoking status at time of delivery - Female - All ages	Note from Data for London: Croydon Observatory provided access to this dataset. TI delivery	Good	Good	Good

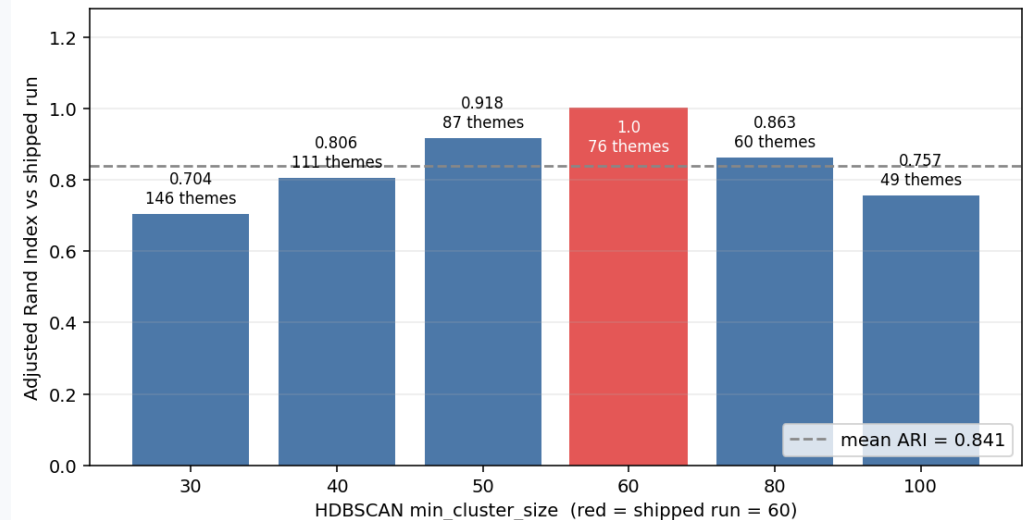
Verdict: tags are good enough to suggest, not good enough to auto-publish. A simple digit-filter removes most of the noise before it reaches a human reviewer.

Do the Themes Make Sense to a Human?

Theme interpretability (A) + sample consistency (B) — HDBSCAN themes

Diabetes / health Clear <i>top words: diabetes, type, received, people</i> 325 datasets · 5 random samples: • People with type 2 diabetes who achieved a blood glu... • People with type 1 diabetes who received urinary alb... • People with type 1 diabetes whose smoking status is ... • Ratio of people with type 1 diabetes who received a ... • People with type 1 diabetes who received a blood pre...	Road traffic & accidents Clear <i>top words: traffic, accidents, road, 24</i> 217 datasets · 5 random samples: • Emergency admissions for pedestrians (aged 0-24) - M... • Motorcyclists killed or seriously injured in road tr... • Pedestrians killed or seriously injured in road traf... • Pedestrians killed or seriously injured in road traf... • Serious casualties from road traffic accidents (aged...	Homelessness & housing Clear <i>top words: homelessness, duty, owed, households</i> 89 datasets · 5 random samples: • Homelessness - households with dependent children ow... • Homelessness - households owed a duty under the Home... • Homelessness - households owed a duty under the Home... • Homeless young people aged 16-24 - Persons - 16-24 y... • Total number of households assessed as not threatene...
Emissions / energy Clear <i>top words: emissions, energy, co2e, gas</i> 277 datasets · 5 random samples: • CH4 emissions from industry gas (kt CO2e) • CO2 within the scope of local authorities from indus... • Total CH4 emissions from commercial sectors (kt CO2e... • Emissions from LULUCF Peatland • Area (km2) (kt CO2e)	Sexual health Clear <i>top words: sexual, sti, chlamydia, hiv</i> 71 datasets · 5 random samples: • Chlamydia detection rate per 100,000 aged 15 to 24 ~... • Gonorrhoea diagnostic rate per 100,000 • Chlamydia detection rate per 100,000 aged 15 to 24 ~... • Chlamydia detection rate per 100,000 aged 15 to 24 ~... • Chlamydia detection rate / 100,000 aged 15-24 - Fema...	Apprenticeships / skills Clear <i>top words: apprenticeships, apprenticeship, starts, number</i> 63 datasets · 5 random samples: • Number of apprenticeship starts as a percentage of t... • Number of apprenticeships achieved/completed as a pe... • Construction, Planning and the Built Environment • Total • Number of apprenticeships achieved/completed as a pe...
Smoking prevalence Clear <i>top words: smoking, 18, prevalence, smokers</i> 152 datasets · 5 random samples: • Smoking Prevalence in adults (18+) - current smokers... • C18 - Smoking Prevalence in adults (18+) - current s... • Smoking Prevalence in adults (18+) - current smokers... • GP patient survey: smoking prevalence - Persons - 16... • C18 - Smoking Prevalence in adults (18+) - current s...	Childhood obesity Clear <i>top words: overweight, obesity, centile, bmi</i> 84 datasets · 5 random samples: • Percentage of adults (aged 18+) classified as overwe... • Year 6: Prevalence of overweight (including obesity)... • Percentage of reception children remaining overweigh... • Year 6: Prevalence of obesity (including severe obes... • Year 6: Prevalence of underweight - Persons - 10-11 ...	Transport / journeys Clear <i>top words: journey, road, transport, time</i> 76 datasets · 5 random samples: • Time (in minutes) by public transport or walking to ... • Time (in minutes) by walking to the nearest secondar... • Time (in minutes) by car to the nearest employment c... • Time (in minutes) by walking to the nearest town cen... • Time (in minutes) by public transport or walking to ...

C Theme stability under parameter change (HDBSCAN) 1 = identical partition, 0 = random



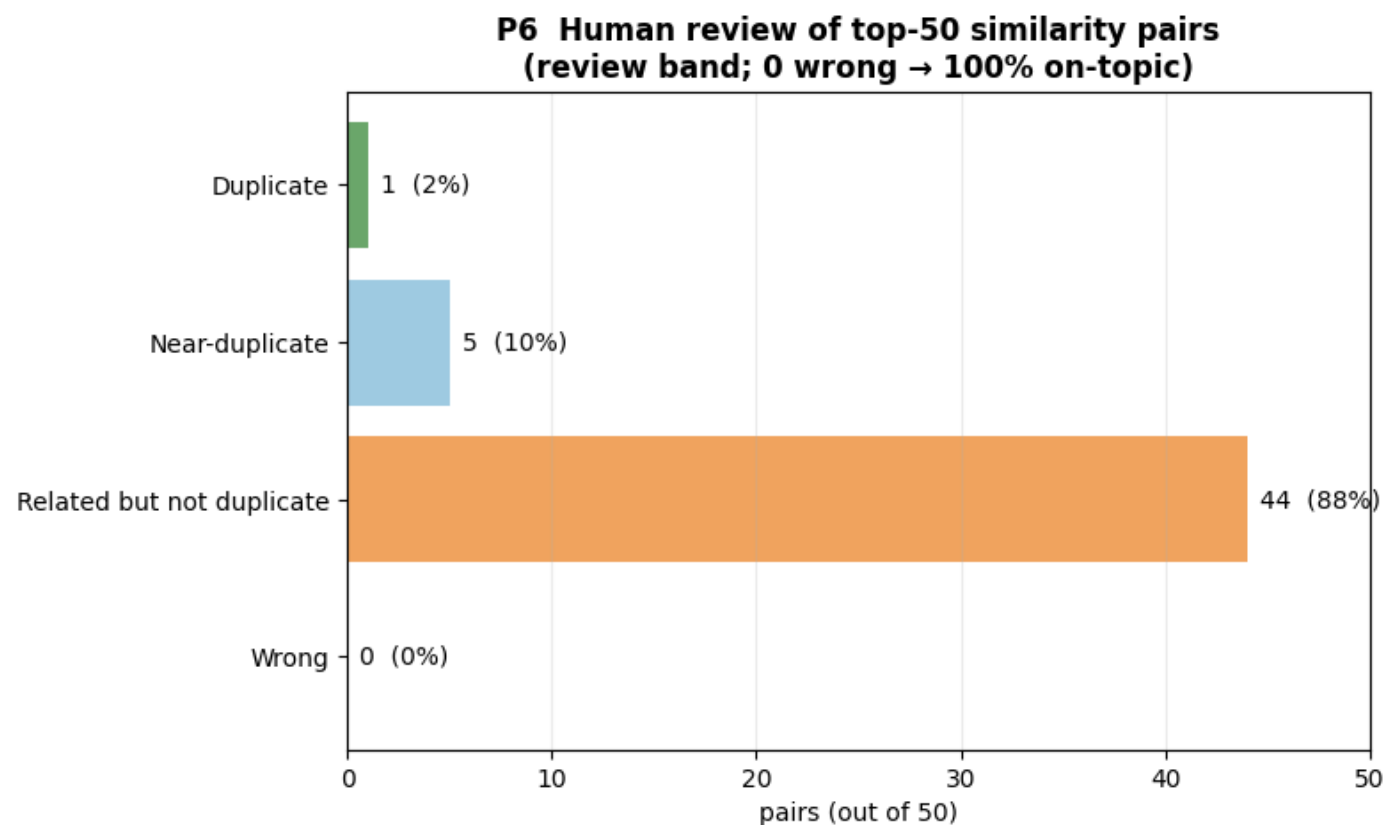
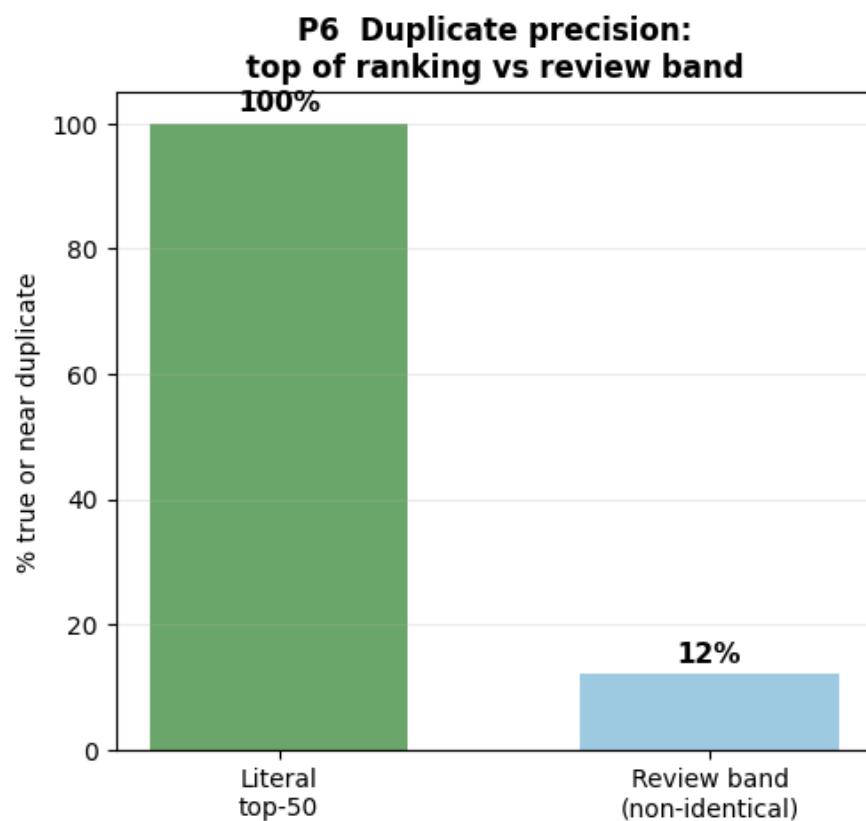
A • Interpretability 71 of 76 themes read as a clear topic

B • Sample consistency 5 random datasets per theme belong together

C • Stability themes hold when parameters change (mean ARI 0.841)

Flagged for Review — Not Auto-Deleted

Near-duplicate detection — flagged for human review, not auto-deleted

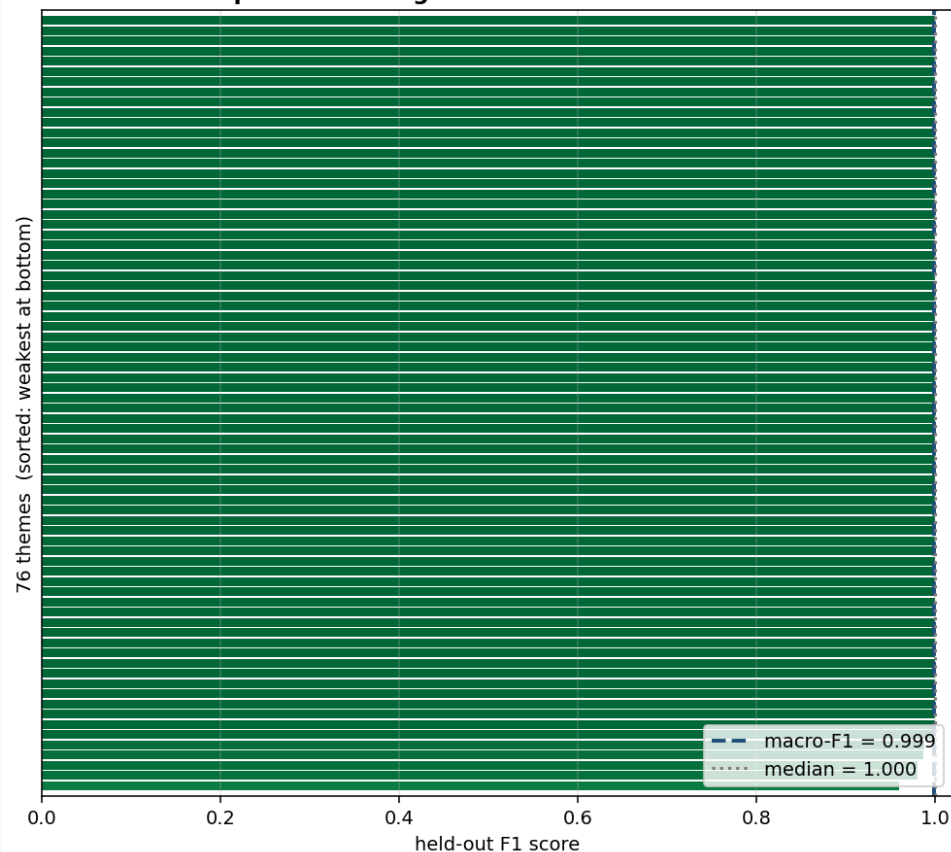


Honest finding: among non-identical high-similarity pairs, only 12% turn out to be true or near duplicates on human review — 88% are just related, not duplicate.

How We Chose — and Checked — Our Method

London Data Week - Theme quality & skore methodological check

Per-theme recovery F1 (held-out 25%)
replaces the illegible 76x76 confusion matrix



skore-style methodological audit

76 themes | macro-F1 = 0.999 | accuracy = 0.999

held-out test set: 3,343 datasets (25%)

skore.train_test_split flagged:

- some themes have <100 datasets in the test set -> scores are noisy
- class sizes are very uneven (theme sizes range widely)
- shuffle=True can inflate scores if data is time-ordered

Weakest themes -> hand-review first:

- F1 0.96 T39: school, pupils, schools, number
- F1 0.98 T41: exception, indicators, sum, except...
- F1 0.98 T17: absence, school, pupil, schools
- F1 0.99 T28: religion, ts031, variable, questio...
- F1 0.99 T42: smoking, 18, prevalence, smokers

Honest caveat

F1 proves themes are LEARNABLE / REPRODUCIBLE from the embeddings -- NOT that a theme/tag is the RIGHT one for a dataset. That is a human call: see [human_review_template.csv](#).

What We Trust — and What We Don't

We do not claim the AI-generated metadata is automatically correct.

Can trust

- Search: semantic beats keyword on every IR metric (Precision@5 0.97 vs 0.42) — including a synonym it was never told, NEET
- Tags: 93.3% specific and usable as suggestions
- Themes: stable under parameter change (ARI 0.841); genuinely interpretable samples

Not yet

- Duplicates: only 12% of non-identical high-similarity pairs are true/near duplicates — needs human review, not auto-merge
- Themes: still some source (borough) bias in a subset of clusters
- 35% of datasets are honestly left 'unclustered' — real metadata gaps, not hidden

Thank you.

Trustworthy as a decision-support layer — not an unchecked replacement for human metadata curation.